Syed Ali Muqtadir

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Education

Oct 2022 – Present
PhD in Neuroscience — University of Stirling, United Kingdom
Thesis title: The Heading Brain: Characterising the Acute Neurobiological Response to Football Heading
Advisors: Dr Magdalena Ietswaart, Dr Colin Moran, Prof Lindsay Wilson
Submitted: 30 September 2026

Sep 2017 – Dec 2018
MSc in Neuroscience — King's College London, United Kingdom
Thesis title: Effect of Transcranial Direct Current Stimulation on Probabilistic Learning in Patients with Schizophrenia
Advisor: Dr Natasza Orlov

Sep 2012 – Dec 2016
BSc (Hons) in Psychology — University of Central Punjab, Pakistan

Research Experience

Oct 2022 – Sep 2026
Doctoral Researcher — University of Stirling, United Kingdom

- Completed a Medical Research Scotland-funded PhD project comprising three studies characterising the acute neurobiological response to controlled football heading.
- Collected data from TMS-EEG, EEG/ERP and cognitive tests from 25 participants before and after heading and control exposure to index cortical E-I balance.
- Acquired and analysed 7 T MRS data before and after heading at ICE, Glasgow and Forschungszentrum Jülich.
- Investigated acute trajectory changes in plasma biomarkers of neuronal and glial injury (NfL, GFAP, UCH-L1, BD-tau) following heading exposure in a 24-hour serial blood-sampling protocol (n = 12, 10 time points) using ultra-sensitive immunoassays (Simoa) at the UK DRI, Edinburgh.
- Initiated a collaboration with the School of Engineering, Edinburgh, supplying heading kinematic data for computational simulations of brain biomechanics (EdiFEHM). Collaborated with industry partner Magstim to test single-pulse and paired-pulse TMS protocols for the project.
- Contributed to increasing technical capabilities at Stirling with the first use of TMS-EEG in the department and implementation of a low-cost navigated TMS system for coil and head tracking.
- Secured additional funding through two successful grant applications. Managed ethics and governance approvals at three sites. Acquired certifications in phlebotomy, IV cannulation and MR safety.

Aug 2019 – Aug 2022
Research Associate — Lahore University of Management Sciences, Pakistan

- Contributed to three research projects on online misinformation, social media communication and digital content virality.
- Applied a logistic diffusion model to engagement data for 'viral' YouTube videos from a large public dataset to identify virality tipping points (Bhimani et al., 2024).
- Collaborated on drafting a manuscript for a study on how the spread of anti-vaccine misinformation is modulated by certain cognitive heuristics (Ali et al., 2022).

Feb 2018 – Aug 2018
Graduate Researcher — King's College London, United Kingdom

- Investigated the effects of tDCS on cognitive function in patients with schizophrenia. Worked within the South London and Maudsley NHS Foundation Trust to administer tDCS protocols, PANSS interviews and cognitive assessments. This work resulted in two shared-first author publications (Scientific Reports, 2021; Psychiatry Research, 2022).

Outputs

Authors who contributed equally are marked with a †.

Publications

1. Bhimani, A., Zainulabdin, K., Ali, K., **Ali Muqtadir, S.**, & Hausken, K. (2024). Social media virality: Reaching the tipping point. *Journal of International Business Research and Marketing*, 8(3), 27–41. https://doi.org/10.18775/jibrm.1849-8558.2015.83.3003
2. Ali, K., Li, C., Zain-ul-abdin, K., & **Muqtadir, S. A.** (2022). The effects of emotions, individual attitudes towards vaccination, and social endorsements on perceived fake news credibility and sharing motivations. *Computers in Human Behavior*, 134, 107307. https://doi.org/10.1016/j.chb.2022.107307
3. Orlov, N. D.†, **Muqtadir, S. A.**†, Oroojeni, H., Averbeck, B., Rothwell, J., & Shergill, S. S. (2022). Stimulating learning: A functional MRI and behavioral investigation of the effects of transcranial direct current stimulation on stochastic learning in schizophrenia. *Psychiatry Research*, 317, 114908. https://doi.org/10.1016/j.psychres.2022.114908
4. Orlov, N. D.†, Sanderson, J.†, **Muqtadir, S. A.**†, Kalpakidou, A. K., Michalopoulou, P. G., Lu, J., & Shergill, S. S. (2021). The effect of training intensity on implicit learning rates in schizophrenia. *Scientific Reports*, 11(1), 6511. https://doi.org/10.1038/s41598-021-85686-5

Preprints

5. **Muqtadir, S. A.**, Aston, T., Reid, K. R., Lember, L.-M., Di Virgilio, T. G., Keith, G., Hunt, D., Teixeira-Dias, F., Binkofski, F., Moran, C. N., & Ietswaart, M. (2026). Acute temporal dynamics of brain injury plasma biomarkers following controlled football heading. *bioRxiv*. https://doi.org/10.64898/2026.09.09.749919

In preparation

6. Patel, H. J.†, **Muqtadir, S. A.**†, Choi, C.-H., Shaikh, U. J., Oros-Peusquens, A.-M., Shah, N. J., Moran, C. N., Ietswaart, M., & Binkofski, F. Neurometabolite changes following football heading: A 7 T magnetic resonance spectroscopy study.

Preregistrations

- **Muqtadir, S. A.**, Aston, T., Moran, C., & Ietswaart, M. (2025). The effect of football heading on blood biomarkers of brain injury. *ISRCTN*. https://doi.org/10.1186/ISRCTN44241334
- **Muqtadir, S. A.** (2025). Impact of football heading on TMS-EEG indicators of neural excitation and inhibition. *OSF*. https://doi.org/10.17605/OSF.IO/4AC6J

Funding and Awards

2022 – 2026	PhD Studentship	Medical Research Scotland
2025	Facility Access Funding Grant	RS MacDonald Charitable Trust
2022	Early Career Exchange Fund	SINAPSE (Scottish Imaging Network)
2013 – 2016	Academic Merit Scholarship	University of Central Punjab

Skills

Brain stimulation	TMS (single-pulse; paired-pulse SICI, LICI, ICF), concurrent TMS-EEG (TEP analysis), tDCS
Neuroimaging	7 T MRS (including edited MRS), structural MRI (volumetric analysis), EEG/ERP (acquisition, preprocessing, spectral analysis)
Physiology	EMG (motor-evoked potentials, cortical silent period), venepuncture and peripheral IV cannulation (certified), eye-tracking (Gazepoint), Near Point Convergence
Biological	Blood sample processing (plasma separation and storage), plasma biomarker quantification (Quanterix HD-X)
Assessment	Structured clinical interviews (PANSS), cognitive and neuropsychological assessments (CANTAB, Cogstate, WASI)
Programming & software	R (statistical analysis, data visualisation), MATLAB (EEGLAB, Osprey, SPM), FSL, FreeSurfer, BrainVision Analyzer, Biopac AcqKnowledge

Teaching and Supervision

University of Stirling

Lecturing

- Delivered teaching for a final-year undergraduate module, "Advanced Methods in Human Physiology Research", on using Transcranial Magnetic Stimulation (TMS) in research, including a practical demonstration.

Project and dissertation supervision

- Co-supervised a master's project investigating EEG activity patterns and natural environment exposure, providing training in EEG data collection and analysis.
- Supervised a third-year group research project, guiding students through the research pipeline from design and ethical considerations to data analysis and writing.

Tutoring and small-group teaching

- Tutored final-year undergraduate students in workshops on writing study proposals based on real-world scenarios.

Laboratory-based teaching

- Led and supported practical laboratory sessions on TMS and EEG for psychology master's students.
- Delivered a five-week training series on EEG/ERP analysis in BrainVision Analyzer for work-placement students and staff.
- Supervised practical fat pad biopsy labs for undergraduates

Assessment

- Marked a variety of assignments across several undergraduate modules, including Introductory Psychology, Developmental Psychology and Psychology in Context.

Lahore University of Management Sciences

- Trained undergraduate researchers in eye-tracking data collection using a Gazepoint system.
- Delivered teaching on research methods for undergraduate students.

Presentations and Public Engagement

Mar 2026 — Talk on 'What does football heading do to the brain?' for Scottish schools — MRS Academy

Mar 2024 — Poster presentation: "Investigating the Acute Effects of Repeated Subconcussive Head Impacts with Football Heading" — Medical Research Scotland RDE

Courses and Certifications

Jul 2026 — MRI Safety Training for MHRA Category B Staff — NHS Scotland TURAS Learn

Mar 2025 — Peripheral IV Cannulation — Geopace Training

Sep 2023 — Visceral Mind: Functional Neuroanatomy Summer School — Bangor University

Jun 2023 — Phlebotomy — Geopace Training